Skills Summary

Rounding Boundaries and Place-Holding Zeros

Use this reference while completing your worksheet or when studying.

Skill 1 — Round to a place you are given

Three steps:

- Underline the digit in the place you are rounding to.
- Look at the **one digit to its right**. That digit decides, and no other digit does.
- 5 or more: the underlined digit goes up by one. Less than 5: it stays. Everything to the right becomes 0.

Example: Round 6,382 to the nearest hundred.
Step 1. The hundreds digit is 3, so the digit that decides is the tens digit just to its right.
Step 2. The tens digit is 8, which is 5 or more, so the hundreds go up: **6,400**.

Try it: Round 7,418 to the nearest hundred.

check: **7,400**

Skill 2 — The exact halfway case

Rule: when the deciding digit is exactly 5, round **up**. “5 or more” includes the 5 itself, so 3,650 rounds to 3,700, not 3,600.

And round the original number only. Rounding to the ten first and then to the hundred can push an answer past the halfway point that never should have crossed it.

Example: Round 3,650 to the nearest hundred.
Step 1. The deciding digit is the tens digit, and it is exactly 5 — the halfway case, which the rule sends up.
Step 2. The hundreds go from 6 to 7 and the smaller places become zeros: **3,700**.

Try it: Round 2,450 to the nearest hundred.

check: **2,500**

Skill 3 — Zeros hold the places open

Rule: a rounded number has the same number of places it started with. 5,048 to the nearest hundred is 5,000, never 50 — the zeros keep the 5 in the thousands place.

The same idea handles a rollover: when a place fills up to 10, it becomes 0 and one unit carries into the place on its left.

Example: What is 10 more than 3,995?

Step 1. Adding a ten fills the tens place past 9, so expect a carry rather than a single changed digit.

Step 2. 9 tens $+1$ ten $=$ 10 tens $=$ 1 hundred, and 9 hundreds $+1$ hundred $=$ 1 thousand, giving **4,005**.

Try it: What is 100 more than 4,950?

check: **5,050**